

In agritech, a singular “all-in-one” AI agent readily provides contradictory guidance, contravenes data regulations, and functions under insufficient supervision. A team of skilled agents, each limited by RANK, TRAIL, HUNT, GUARD, and CYCLE, enables AgriPride to assist farmers with safer, clearer, and more useful automation.

Agent 1 – Scout Agent (Market Intelligence)

The price advisor provides farmers with current crop price recommendations and financial knowledge tips via SMS but doesn't make trades or adjust orders.

An authority limit of 3 SMS per farmer each day.

Farmers always make the ultimate judgment; agreements aren't locked in.

Notice automatically alerts Guardian Logistics Agent and AgriPride supervisor of “high-risk pricing environment” if agricultural market price fluctuation surpasses 20% in 24 hours.

Kill Switch:

AgriPride support workers can immediately suspend all Scout messages to a region or farmer group using USSD code *#700# amid suspected system problems or regulatory concerns.

One guardrail: Pricing guidance must use market and quality indicators, not gender, ethnicity, or village name as a risk proxy.

One CYCLE part:

Capture: Track whether each farmer followed advice (“sell now” vs. “wait”) and the price they got to quantify advice quality over time.

One particular tool, LangGraph, can be used to visualize the Scout's decision flow (data fetch, volatility check, SMS generation, notification triggers).

3. Agent 2 – Guardian Agent (Logistics Coordinator)

Here, you implement TRAIL, one GUARD, one CYCLE, and a tool.

Utilize precisely the TRAIL slots in the summary: Ephemeral memory (T)

Retains solely the current booking request: farmer ID, crop kind, volume, pickup date, and provisional route for that particular transaction. Cleared upon completion or cancellation. Relational memory (R) upon explicit consent from the farmer, maintains consistent information, including regular farm location, desired collection place, and average harvest period, and is utilized to pre-populate future booking forms.

Archival memory (A) maintains anonymized historical route success ratings (e.g., “Route X: 82% on-time deliveries”) devoid of phone numbers or names, to enhance route selection and truck allocation.

Principles of Inheritance (I)

Only confirmed road condition data and route risk scores are transmitted to the Hunter Agent, excluding raw farmer identification; the Hunter perceives “Route A: flood risk high” instead of actual phone numbers.

Land Rights (L)

All logistics data is housed on AWS Africa (Cape Town) or a comparable location, adhering to Uganda's Data Protection Act and Rwanda's data residency requirements; personal data will not be sent outside of Africa.

One GUARD safety railing

Guardrail: The Guardian Agent is prohibited from confirming any truck booking on a route designated as "unsafe" (e.g., conflict, landslides) within the past 7 days without human authorization.

One CYCLE element

Monitor the percentage of delayed deliveries and their key causes (road conditions, truck breakdowns, double-booking), providing weekly analyses for logistics enhancements.

A singular specialist instrument

Utilize n8n to integrate Guardian with external services like as road condition APIs, SMS gateways, and internal booking systems through a visual process.

4. Agent 3 – Hunter Agent (Human-in-the-Loop Liaison)

Utilize HUNT in conjunction with one GUARD, one CYCLE, and a tool.

Clearly delineate the handoffs:

Handoff triggers (H)

Upon Scout logging a verified buyer contract exceeding 500 kg, the Guardian is activated with the destination, volume, and deadline details.

Upon the successful reservation of a truck by Guardian, Hunter is prompted to create a briefing for the human logistics manager.

Unified context (U)

Hunter accesses a common context object that includes: crop kind, volume, price, route risk score from Guardian, and farmer contact channel (SMS/USSD). There are no unprocessed phone numbers in the shared board.

Regulations of Negotiation (N)

If Guardian indicates a "high route risk" or if freight costs are above 30% of the contract price, Hunter is required to escalate the matter to the personal operations manager for a conclusive decision; autonomous confirmation is prohibited.

Conditions for termination (T):

The workflow concludes when the human manager either approves or alters the strategy.

Hunter transmits a human-validated summary via SMS/USSD to the farmer with the pickup schedule and risk advisories.

One GUARD safety railing

Guardrail: The hunter is required to produce explanations in simple language appropriate for low-literacy farmers, avoiding technical jargon and fear-inducing language; all communications must be evaluated against a dignity standard.

One CYCLE element,

Yield Insights: Weekly, Hunter compiles patterns from escalations (e.g., “30% of deals escalated due to elevated freight costs on Route X”) to inform route redesign and collaboration strategies.

A singular specialist instrument

Utilize CrewAI to oversee role interactions among Scout, Guardian, and Hunter, addressing issues arising from the tension between price optimization and safety or human intervention.

Although AgriPride functions independently, human authority is maintained at all essential junctures. The Scout Agent may provide counsel but is incapable of executing trades; the farmer's decision is conclusive. The Guardian Agent is capable of suggesting paths but cannot bypass stringent safety protocols without human intervention. The primary objective of the Hunter Agent is to inform a human logistics manager, rather than supplant them. TRAIL safeguards agricultural data and retains it inside African countries, whereas CYCLE employs outcomes to enhance suggestions through human validation. Collectively, these design decisions guarantee that AI enhances farmer autonomy instead of undermining it.